

Am I correct that the next step would be to set $\beta = 0$ and solve for $\bar{a}$ and $\bar{b}$?

If we are writing a female's phenotype as $y = a + bn$, hence $\bar{y} = \bar{a} + \bar{b}n$ are we assuming that y scales linearly with n ?

Given the optimum for y doesn't scale linearly with n (below), this seems likely to be violated and I wondered if there is a way around it?

Optimal lay date with c and $\alpha_f = 1$:

